



ISHFAQ AHMAD DAR

AI / ML · Python · Backend

CONTACT

EMAIL

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LOCATION

Kashmir, India

LINKEDIN

ishfaq-ahmad-dar

GITHUB (AI/ML)

github.com/Dar-Ishfaq-1

GITHUB (BACKEND)

Dar-Ishfaq-Backend

PORTFOLIO

ai-and-ml-portfolio.vercel.app

AI / ML SKILLS

- Python (Expert)
- PyTorch · TensorFlow
- Scikit-Learn · XGBoost
- Vision Transformers (ViT)
- CNNs · Grad-CAM · XAI
- FAISS Similarity Search
- OpenCV · Computer Vision
- Hugging Face · LLM APIs
- Pandas · NumPy · EDA
- Streamlit · Gradio

BACKEND & DEV

- FastAPI · Flask · Django
- REST API Development
- Backend Automation
- React · Tailwind CSS
- SQL · SQLite · PostgreSQL
- Docker · Git · GitHub
- AWS (Cloud Practitioner)

EDUCATION

MSc Information Technology
University of Kashmir · 2023–2025
BSc Non-Medical
Govt. Degree College · 2019–2022

CERTIFICATIONS

AWS Cloud Practitioner
Data Analytics — TuteDude
Power BI & SQL — Simplilearn
Python AI Dev — Mimo · 2026
Accenture Data Analytics

LANGUAGES

English — Professional
Urdu — Native
Kashmiri — Native

STATUS

● Open to Work

AI/ML · Backend · Python Dev

PROFESSIONAL SUMMARY

Versatile AI/ML Engineer and Python Developer with an MSc in IT, combining deep expertise in computer vision, deep learning, and explainable AI with strong backend engineering across FastAPI, Flask, Django, and REST APIs. Specialist in Vision Transformers (ViT), FAISS similarity search, and Grad-CAM interpretability — delivering 92% Top-5 retrieval accuracy on medical imaging. AWS Certified, Docker-proficient, and experienced building full-stack AI systems from model training to production API deployment. Targeting AI/ML Engineer, Python Developer, Data Science, and Full-Stack AI roles across IT Services, Software Development, and Artificial Intelligence domains.

AI / ML PROJECTS

Content-Based Medical Image Retrieval (CBMIR) *[Flagship]*

PyTorch · ViT · FAISS · Grad-CAM · Gradio · OpenCV · Python · XAI

- Built an explainable CBIR engine using ViT embeddings + FAISS ANN indexing, achieving 92% Top-5 retrieval accuracy on medical imaging benchmarks with sub-400ms query latency.
- Implemented full pipeline: preprocessing → ViT feature extraction → FAISS cosine indexing → ranked retrieval → Grad-CAM heatmap overlay for clinically auditable decisions.
- Deployed interactive Gradio demo for real-time image querying with side-by-side XAI visualization, suitable for clinical workflow integration.

Text-to-SQL Agent — NL → SQL via Gemini AI

Python · Flask · Google Gemini API · SQLite · React · Tailwind CSS · REST API · Docker

- Architected a full-stack AI app converting natural language to validated SQL using Gemini with schema-aware prompting and SELECT-only safety enforcement to prevent injection.
- Built Flask RESTful backend (/api/query, /api/schema, /api/health) + React/Tailwind frontend; containerised with Docker for reproducible cloud deployment.

Weather Analytics & Forecasting Platform

Python · XGBoost · Prophet · Scikit-Learn · Streamlit · SQLite · Docker · APScheduler

- Built an ML forecasting platform with 24-hour and 7-day predictions using Random Forest, Prophet, and XGBoost (MAE, RMSE, R² evaluated) across 20+ cities.
- Implemented Isolation Forest anomaly detection with 15-minute auto-refresh via APScheduler; deployed Streamlit dashboard on Streamlit Community Cloud.

Unsupervised Object Classification — VGG16 + K-Means

PyTorch · VGG16 · Scikit-Learn · PCA · K-Means · NumPy · Matplotlib

- Zero-label clustering pipeline using VGG16 (ImageNet pre-trained) as 512-D feature extractor; PCA to 61D + K-Means (k=5) achieved Silhouette Score 0.71 on flower species.

BACKEND & PYTHON PROJECTS

REST API Development & Backend Automation Suite

Python · FastAPI · Django · Flask · Docker · OOP · Git

- Delivered multiple REST APIs using FastAPI and Django with modular OOP architecture; automated backend workflows (data ingestion, reporting, notifications) for clients.
- Integrated third-party APIs, enforced input validation pipelines, and deployed containerised applications — clean, well-documented code following industry standards.

Python Backend Portfolio & Contact Form System

Python · Flask · Formsprees · Input Validation · HTML/CSS · Docker

- Built FastAPI-backed portfolio with Docker containerisation and optimised REST endpoints; implemented server-side validation pipeline with structured error handling and fallback email logic.

EXPERIENCE

Python & Backend Developer · [Freelance](#) · [Remote](#) · [Recent](#)

- Delivered Python web applications and REST APIs to real-world clients using FastAPI, Flask, and Django with clean OOP architecture and documented codebases.
- Integrated third-party APIs and automated backend workflows; debugged complex issues systematically and delivered scalable solutions within agreed timelines.

RESEARCH INTERESTS

Computer Vision · Vision Transformers · Explainable AI · Medical AI · Retrieval-Augmented

